**2** **(a)** Explain what is meant by the internal energy of a system.

.....

.....

..... [2]

**(b)** Dry air is enclosed in an air-tight cylinder fitted with a piston, as shown in Fig. 2.1.

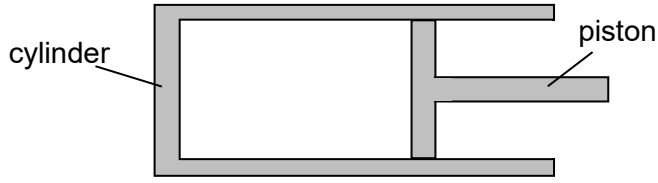


**Fig. 2.1**

The piston moves to compress the air and the variation with volume $V$ of the pressure $p$ of the air during the process from A to B is shown in Fig. 2.2.

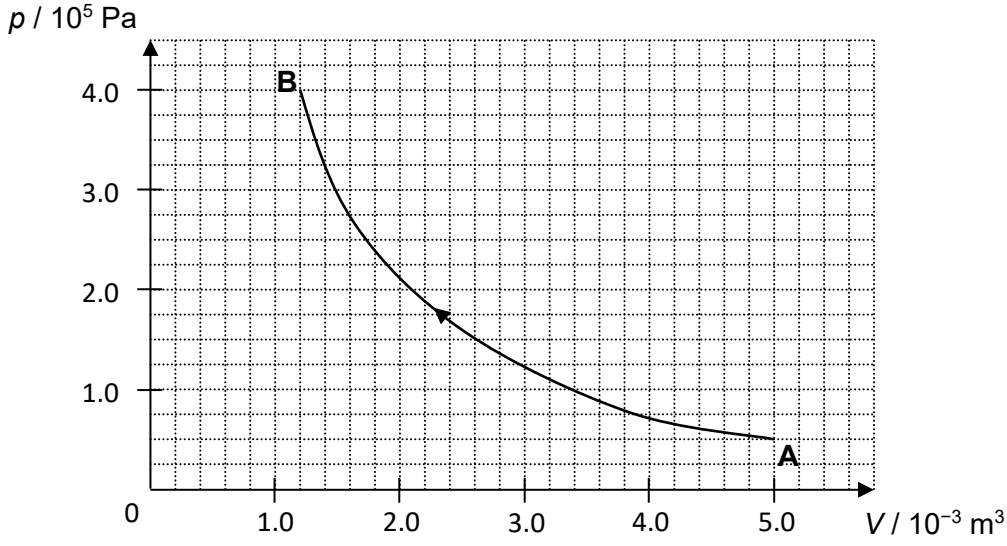


**Fig. 2.2**

It may be assumed that the dry air behaves as an ideal gas.

**(i)** Assume the dry air is a monoatomic gas. Calculate the internal energy of the dry air just before the start of process from A to B.

internal energy = ..... J [2]

**(ii)** The dry air then goes through two more processes.

Process 2: The air is cooled while keeping the piston at the same position.

Process 3: The air then expands, while kept at constant temperature, to return to its original state.

**1.** Calculate the pressure of the air at the end of the process 2.